ACT SUBJECT TEST & DIGITAL SAT MATH COURSES

Rabaa Farrag

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ACT2 • ACT SUBJECT TEST MATH LEVEL 1 • PRACTICE EXAM

ACT2 Exam 7

Full Practice Test • Test #7

Statistics, nets, systems, logic & graphs of functions. Work through all 50 questions under timed conditions, then check your answers with the answer key and the step-by-step solutions at the end.

50	MCQ	60 min
QUESTIONS	MULTIPLE CHOICE	TIME LIMIT

NAME _____ DATE _____ SCORE _____ /50

How to get the most from this practice test

- **Read** every question slowly and underline the key words.
- **Draw** a quick sketch for every geometry question before you solve it.
- **Solve** step by step and show your work.
- **Check** your answer by substituting it back before you choose.

RF

YOUR MATH COACH

Eng. Rabaa Farrag

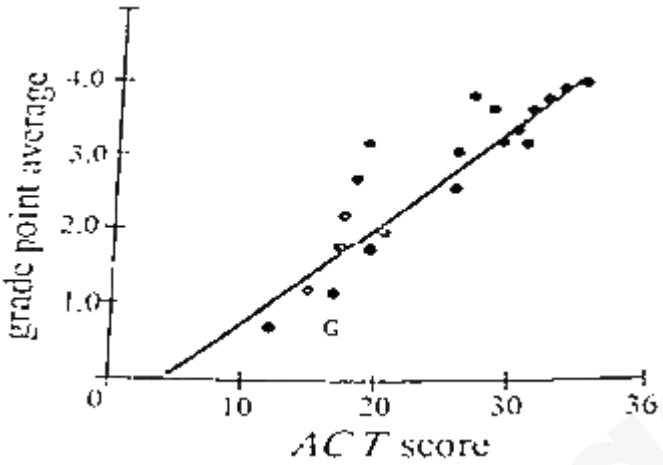
ACT, Digital SAT & EST Math expert with **23 years of experience**, helping students master math and reach **700+**.

ACT SUBJECT TEST • MATH LEVEL 1 • TEST #7

Practice Questions

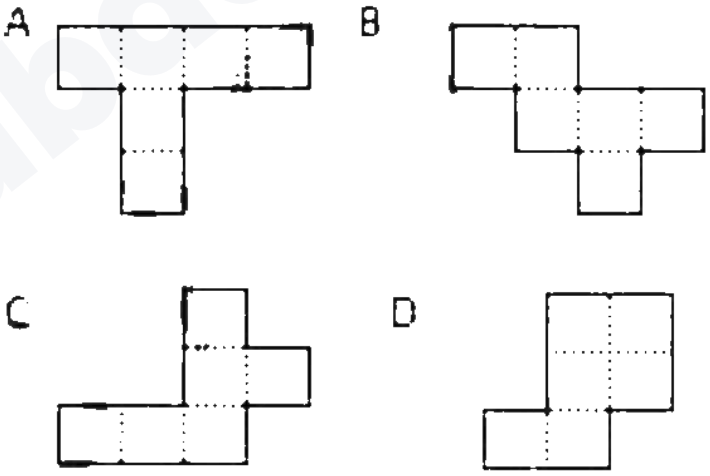
50 QUESTIONS

1 This graph shows the grade point average and ACT score for each student in Mr. Rodriguez's class. Use the line of best fit from the graph to determine which of the following statements must be true for Mr. Rodriguez's class.



(A) There is no correlation between ACT scores and grade point averages.
(B) There is a positive correlation between ACT scores and grade point averages.
(C) As ACT scores increase, grade point averages decrease.
(D) The student with the highest ACT score has the lowest grade point average.

2 When folded along the dashed lines, which of the nets forms a cube?



(A) Net A
(B) Net B
(C) Net C
(D) Net D

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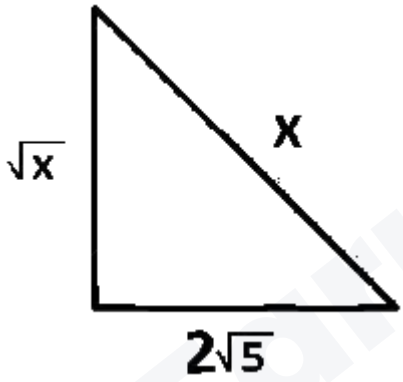
3 Simplify $-4x - 8 - 2(3x + 1) - 10$.

(A) $10x + 20$
(B) $-10x - 17$
(C) $-10x - 20$
(D) $-2x - 20$

4 A list has mean $= 6$, median $= 6$, mode $= 10$. If we wrote a second list with each number increased by 5, the mean of the second list will be:

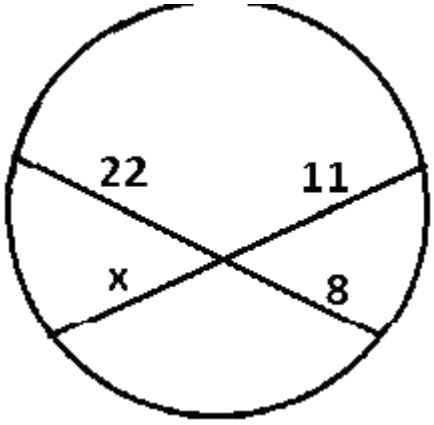
(A) 6
(B) 10
(C) 11
(D) 15

5 What is the value of x?



(A) -4
(B) -5
(C) 4
(D) 5

6 What is the value of x?



(A) 2
(B) 11
(C) 16
(D) 22

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7 What is the area of a circle when the diameter is equal to 17?

(A) 289π
(B) 72.25π
(C) 17π
(D) 34π

8 Triangle DCE is similar to triangle ACB. Which statement is true?

(A) Angle D corresponds to angle B.
(B) Angle E corresponds to angle B.
(C) Angle A corresponds to angle C.
(D) Angle C corresponds to angle B.

9 The points $(2, 4)$ and $(2, -6)$ give which kind of slope?

(A) Positive slope
(B) Negative slope
(C) Undefined slope
(D) Zero slope

10 Simplify $\dfrac{3x^2}{3x^3 + 27x}$.

(A) $\frac{3x^2}{30x^3}$
(B) $\frac{x}{3x^2+3x}$
(C) $\frac{1}{x+9}$
(D) $\frac{x}{x^2+9}$

11 A quadrilateral has interior angles measuring n, $2n$, $3n$, and $4n$. What is the value of n?

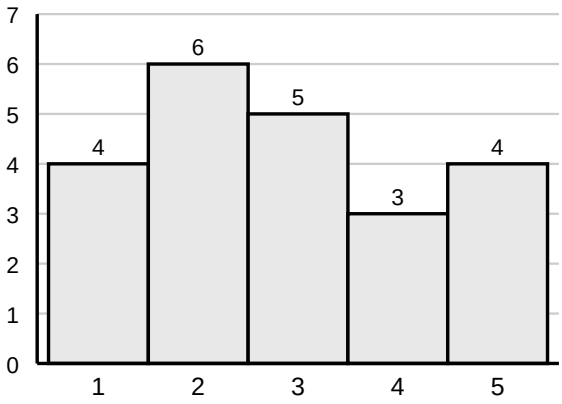
(A) 36
(B) 3.6
(C) 360
(D) 180

12 Find the solution of the following system: $3x + 4 = -4y$ and $2y + x = 2$.

(A) $(-5, 8)$
(B) $(-8, 5)$
(C) $(-10, 4)$
(D) $(10, -4)$

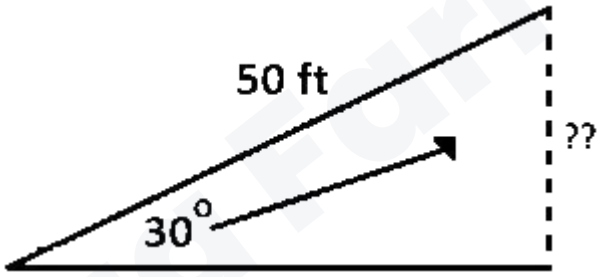
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13 What is the median of the data shown in the bar graph?



(A) 3
(B) 5
(C) 9
(D) 4

14 A bike ramp is 50 ft long, and the angle of elevation is $30°$. To the nearest hundredth of a foot, how high is the end of the ramp?



(A) 25.00
(B) 28.87
(C) 35.36
(D) 43.30

15 Rationalize the denominator and simplify $\frac{30}{\sqrt{20}}$.

(A) 3
(B) $3\sqrt{5}$
(C) $\frac{3\sqrt{2}}{2}$
(D) $3\sqrt{2}$

16 How many points of intersection do the following equations have? $y = \frac{1}{3}x + 5$ and $y = -x^2 + 8$

(A) 0
(B) 1
(C) 2
(D) 3

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17 If $-3x + 7 < 16$, which of the following must be true?

(A) $x > -3$
(B) $x < -3$
(C) $x \geq -3$
(D) $x \leq -3$

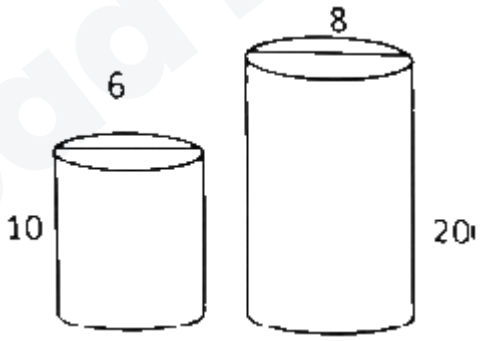
18 Simplify $(5x^3z^4y^{-2})^{-2}(3xy^4z^{-3})^3$.

(A) $\frac{27y^{16}}{25z^{17}x^3}$
(B) $\frac{27y^{16}}{5z^{17}x^3}$
(C) $\frac{3y^{16}}{25z^{17}x^3}$
(D) $\frac{27y^{16}}{25z^2x^3}$

19 Simplify $\frac{\sqrt{a^2b} + \sqrt{ab^2}}{ab}$ for $a, b > 0$.

(A) $\frac{a\sqrt{b}+b\sqrt{a}}{ab}$
(B) $\frac{a^2\sqrt{b}+b^3\sqrt{a}}{ab}$
(C) $\frac{a^3\sqrt{b}+b^2\sqrt{a}}{ab}$
(D) $\frac{a\sqrt{b}+b\sqrt{a}}{\sqrt{ab}}$

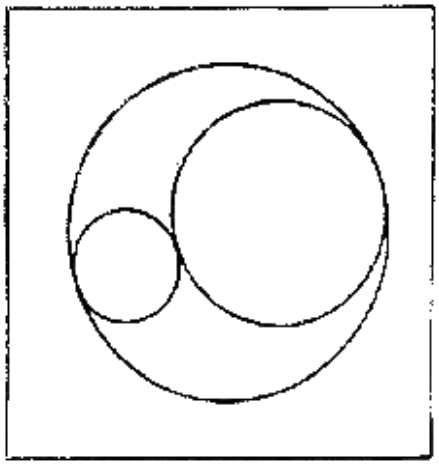
20 To the nearest tenth, how many small cylinders full of water can fit into the large cylinder? (Small: diameter 6, height 10. Large: diameter 8, height 20.)



(A) 3.5
(B) 3.7
(C) 3.6
(D) 3

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21 The diameter of the first circle is 1, the second is 2, and the third (largest) is 3. If a point were to be chosen at random in the square with side length 4, what is the probability of the point landing in the large circle but not in the two smaller tangent circles?



(A) 18%
(B) 19%
(C) 20%
(D) 21%

22 A basketball hoop is 13 ft tall with a shadow of 3 ft 4 in. A boy is 4 ft 5 in tall. What is the length of the shadow he creates?

(A) 1 ft 1.5 in
(B) 1 ft 15 in
(C) 12 ft 1.5 in
(D) 3 ft 1.5 in

23 The mean of 6 numbers is 6, the median is 6, and the mode is 10. If each number is increased by 5, what would be the mean?

(A) 2
(B) 11
(C) 5
(D) 6

24 A line has a y-intercept of 0 and a negative slope. Which quadrants would the line lie in?

(A) Q_1, Q_2
(B) Q_2, Q_4
(C) Q_1, Q_3
(D) Q_3, Q_4

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25 A set of points was reflected across the y-axis. What would be the new set of points after the reflection? $(-3, 5)$, $(-4, 2)$, $(-1, 6)$

A $(3, 5)$, $(4, 2)$, $(1, 6)$
B $(-3, -5)$, $(-4, -2)$, $(-1, -6)$
C $(5, -3)$, $(2, -4)$, $(6, -1)$
D $(3, -5)$, $(4, -2)$, $(-1, 6)$

26 Marco, Charlie, Rob, and Lauren ate desserts; the options were cookies, ice cream, brownies, and cake. Marco and Charlie had the same dessert. No one chose the same dessert as Rob or Lauren. Rob chose brownies and Charlie chose cake. Neither Marco nor Lauren chose cookies. What did Lauren eat?

A Ice cream
B Brownies
C Cookies
D Cake

27 Simplify $\frac{x^2 - 100}{x^2 + 4x - 60} \cdot \frac{x^2 - 6x}{x - 10}$.

A x^2
B x
C $x - 10$
D $x - 6$

28 Find and compare the mean and median of the following numbers:
$75, 70, 80, 70, 80, 90, 95, 70, 70, 75, 80, 80, 90, 95$

A Mean $>$ Median
B Mean $<$ Median
C Mean $=$ Median
D Mean $= 75$, Median $= 85$

29 Chris started a new job. On the first day he worked for 5 hours, on the fifth day he worked for 4 hours, and on the 12th and 14th days he worked for 6 hours. Which relation represents this situation (day, hours)?

A $(1, 5)$, $(5, 4)$, $(12, 6)$, $(14, 6)$
B $(5, 1)$, $(5, 4)$, $(6, 12)$, $(6, 14)$
C $(1, 5)$, $(4, 5)$, $(12, 6)$, $(6, 14)$
D $(1, 5)$, $(5, 4)$, $(6, 12)$, $(14, 6)$

30 If there were 6 rabbits on an isolated island and their number doubles each year, what would be their number after 9 years?

A 3000
B 3070
C 3072
D 300

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31 The radii of 3 concentric circles have a ratio of $5 : 8 : 9$. The circle between the smallest and the largest has a radius of 10 cm. To the nearest hundredth of a centimeter, what is the distance from the smallest circle to the largest circle?

(A) 1
(B) 1.25
(C) 2.5
(D) 5

32 What is the greatest number of squares, 5 cm on a side, that can be cut from a rectangular piece of construction paper 60 cm long and 45 cm wide?

(A) 75
(B) 90
(C) 96
(D) 108

33 A store has square rugs in three sizes. A side of the largest rug is twice as long as a side of the middle one, and a side of the middle one is twice as long as a side of the smallest one. What is the ratio of the area that the largest rug will cover to the area that the smallest one will cover?

(A) 3 to 1
(B) 4 to 1
(C) 8 to 1
(D) 16 to 1

34 If a, b, m and n are integers, $m > 0$, $n < 0$, and $x^2 + mx + n = (x + a)(x + b)$, what must be true of the product and the sum of a and b?

(A) $ab > 0$ and $a + b > 0$
(B) $ab > 0$ and $a + b < 0$
(C) $ab < 0$ and $a + b > 0$
(D) $ab < 0$ and $a + b < 0$

35 When two square regions overlap, what can the shape of the overlapping portion be? I. Square II. Triangular III. Rectangular

(A) I only
(B) II only
(C) III only
(D) I, II, and III

36 A day-care center has a bag of blocks of various geometric shapes. For these blocks, each of these statements is true: If a block is triangular, then it is blue. If a block is not blue, then it is plastic. If a block is square, then it is not blue. If a block is circular, then it is plastic. If a block is not blue, then it is not circular. If a block from the bag is blue, which of the following statements must be true about the block?

(A) It is triangular
(B) It is circular
(C) It is not a square
(D) It is not plastic

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37 Quadrilateral $ABCD$ is inscribed in a circle with a diameter $\overline{AC}$ such that $m\angle DAC = 55°$ and $m\angle BCA = 40°$. What are $m\angle BAD$ and $m\angle BCD$?

A $m\angle BAD = 105°$ and $m\angle BCD = 75°$
B $m\angle BAD = 100°$ and $m\angle BCD = 80°$
C $m\angle BAD = 95°$ and $m\angle BCD = 85°$
D $m\angle BAD = 95°$ and $m\angle BCD = 90°$

38 Point P is on the line $y = 3$. The distance from P to the point with coordinates $(8, 8)$ is 13 units. What is the x-coordinate of P if it is less than 8?

A -5
B -4
C 0
D 3

39 What is the slope of the line whose equation is $5x - 3y = 9$?

A -3
B $-\frac{5}{3}$
C $\frac{5}{3}$
D 5

40 If $5 \times (P + Q) = 30$ and P is greater than zero, then Q could NOT be:

A 7
B 4
C $3\frac{1}{2}$
D 0

41 What property of real numbers is demonstrated by the statement "If $a = 20$, then $20 = a$"?

A Transitive Property of Equality
B Substitution Property of Equality
C Reflexive Property of Equality
D Symmetric Property of Equality

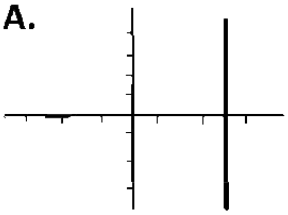
42 Which statements are always logically equivalent?

A The converse of a conditional statement and the contrapositive of a conditional statement
B The conditional statement and its inverse
C The conditional statement and its contrapositive
D The inverse and contrapositive of a conditional statement

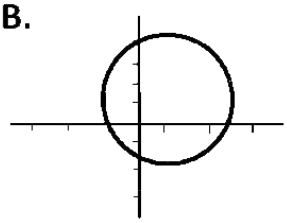
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43 Which graph shows a function where x is the input and y is the output?

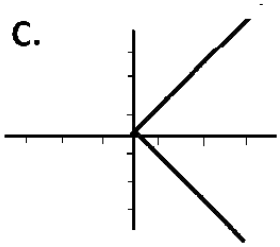
A.



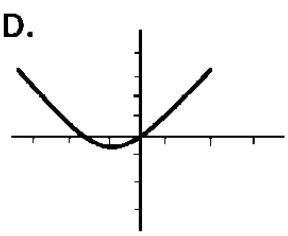
B.



C.

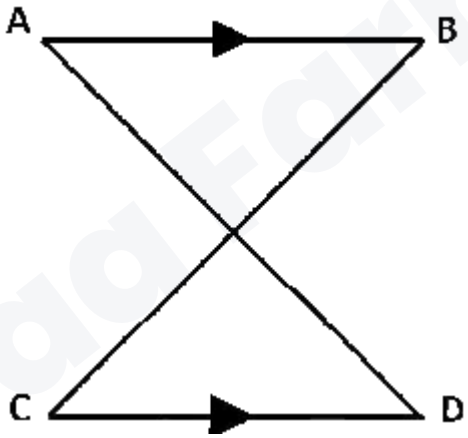


D.



(A) Graph A
(B) Graph B
(C) Graph C
(D) Graph D

44 Which of the following proves that the two triangles are similar?



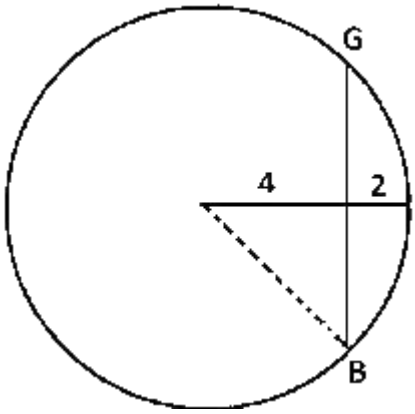
(A) AA
(B) ASA
(C) SAS
(D) SSS

45 Which is the converse of the following statement? "If Chris uses toothpaste X, he doesn't get cavities."

(A) If Chris doesn't get cavities, he uses toothpaste X.
(B) If Chris gets cavities, he uses toothpaste X.
(C) If Chris doesn't get cavities, he doesn't use toothpaste X.
(D) If Chris doesn't use toothpaste X, he doesn't get cavities.

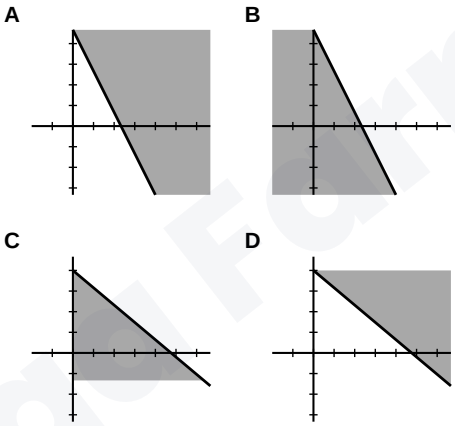
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46 $\overline{GB}$ is a chord. Find the value of GB.



(A) $2\sqrt{5}$
(B) $4\sqrt{5}$
(C) $\sqrt{5}$
(D) $2\sqrt{4}$

47 A represents adults while S represents children: $6A + 3S \geq 1500$. Which graph should be used?

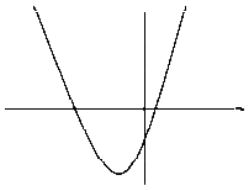


(A) Graph A
(B) Graph B
(C) Graph C
(D) Graph D

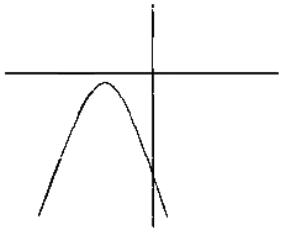
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48 You have the quadratic $y = x^2 + bx + c$ with $b > 0$ and $c < 0$. Which graph could it be?

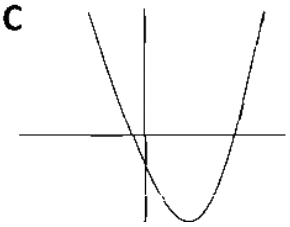
A



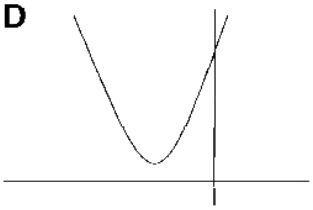
B



C



D



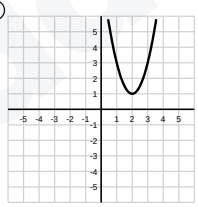
(A) Graph A
(B) Graph B
(C) Graph C
(D) Graph D

49 Determine the number of solutions for the following system of equations: $2x + 5y = 7$ and $10y = -4x + 14$.

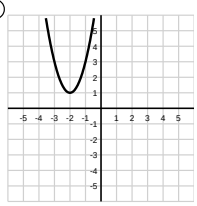
(A) Exactly one solution
(B) No solutions
(C) Infinitely many solutions
(D) Exactly two solutions

50 Which of these graphs shows the quadratic function $y = 2x^2 + 8x + 7$?

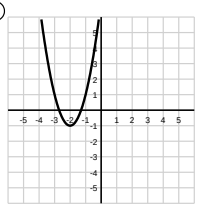
(A)



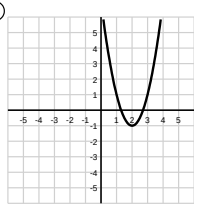
(B)



(C)



(D)



(A) Graph A
(B) Graph B
(C) Graph C
(D) Graph D

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TEST #7 • CHECK YOUR WORK

Answer Key

50 ANSWERS

Q	ANSWER	Q	ANSWER
1	B	26	A
2	B	27	B
3	C	28	C
4	C	29	A
5	D	30	C
6	C	31	D
7	B	32	D
8	B	33	D
9	C	34	C
10	D	35	D
11	A	36	C
12	B	37	A
13	A	38	B
14	A	39	C
15	B	40	A
16	C	41	D
17	A	42	C
18	A	43	D
19	A	44	A
20	C	45	A
21	C	46	B
22	A	47	D
23	B	48	A
24	B	49	C
25	A	50	C

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TEST #7 • STEP-BY-STEP

Detailed Solutions

50 SOLUTIONS

1 **Answer: B** There is a positive correlation between ACT scores and grade point averages.

Step 1: The line of best fit rises from left to right.
Step 2: As ACT scores increase, GPAs tend to increase: a positive correlation.
Step 3: ("Must be true" rules out claims about a single student.)

2 **Answer: B** Net B

Step 1: A cube net needs 6 squares arranged so that no two faces overlap when folded.
Step 2: Net B (a staircase of 1–3–2 squares) folds so that each square becomes a different face.
Step 3: The other nets have overlapping faces when folded.

3 **Answer: C** $-10x - 20$

Step 1: Distribute: $-2(3x + 1) = -6x - 2$.
Step 2: $-4x - 8 - 6x - 2 - 10$.
Step 3: $= -10x - 20$.

4 **Answer: C** 11

Step 1: Adding 5 to every number in a list adds 5 to the mean.
Step 2: New mean $= 6 + 5 = 11$.

5 **Answer: D** 5

Step 1: Right triangle with legs $\sqrt{x}$ and $2\sqrt{5}$, hypotenuse x: $(\sqrt{x})^2 + (2\sqrt{5})^2 = x^2$.
Step 2: $x + 20 = x^2 \Rightarrow x^2 - x - 20 = 0 \Rightarrow (x - 5)(x + 4) = 0$.
Step 3: A length is positive: $x = 5$.

6 **Answer: C** 16

Step 1: Intersecting chords: the products of the two segments of each chord are equal.
Step 2: $22 \times 8 = 11 \times x$.
Step 3: $176 = 11x \Rightarrow x = 16$.

7 **Answer: B** 72.25π

Step 1: $r = \frac{17}{2} = 8.5$.
Step 2: $A = \pi r^2 = 72.25\pi$.

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8 **Answer: B** Angle E corresponds to angle B.

Step 1: The order of the letters gives the correspondence: $D \leftrightarrow A$, $C \leftrightarrow C$, $E \leftrightarrow B$.
Step 2: So angle E corresponds to angle B.

9 **Answer: C** Undefined slope

Step 1: Slope $= \frac{-6-4}{2-2} = \frac{-10}{0}$ — division by zero.
Step 2: Both points have $x = 2$: a vertical line, so the slope is undefined.

10 **Answer: D** $\frac{x}{x^2+9}$

Step 1: Factor the denominator: $3x^3 + 27x = 3x(x^2 + 9)$.
Step 2: $\frac{3x^2}{3x(x^2+9)} = \frac{x}{x^2+9}$.

11 **Answer: A** 36

Step 1: The angles of a quadrilateral add to $360°$.
Step 2: $n + 2n + 3n + 4n = 10n = 360$.
Step 3: $n = 36$.

12 **Answer: B** $(-8, 5)$

Step 1: From the second equation: $x = 2 - 2y$.
Step 2: Substitute: $3(2 - 2y) + 4 = -4y \Rightarrow 10 - 6y = -4y \Rightarrow y = 5$.
Step 3: $x = 2 - 10 = -8$. Solution $(-8, 5)$.

13 **Answer: A** 3

Step 1: Read the bars as how often each value 1–5 occurs: $4, 6, 5, 3, 4$ (total 22).
Step 2: The median is the average of the 11th and 12th values. Cumulative: value $1 \to 4$, value $2 \to 10$, value $3 \to 15$.
Step 3: Both the 11th and 12th values are 3, so the median is 3.

14 **Answer: A** 25.00

Step 1: The ramp is the hypotenuse (50 ft) and the height is opposite the $30°$ angle.
Step 2: $h = 50 \sin 30° = 50 \times 0.5 = 25.00$ ft.

15 **Answer: B** $3\sqrt{5}$

Step 1: $\sqrt{20} = 2\sqrt{5}$, so $\frac{30}{2\sqrt{5}} = \frac{15}{\sqrt{5}}$.
Step 2: $\frac{15\sqrt{5}}{5} = 3\sqrt{5}$.

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16 **Answer: C** 2

Step 1: Set equal: $-x^2+8=\frac{1}{3}x+5 \Rightarrow x^2+\frac{1}{3}x-3=0$.
Step 2: Discriminant $=\frac{1}{9}+12>0$: two real solutions.
Step 3: 2 points of intersection.

17 **Answer: A** $x>-3$

Step 1: Subtract 7: $-3x<9$.
Step 2: Divide by -3 and flip the sign: $x>-3$.

18 **Answer: A** $\frac{27y^{16}}{25z^{17}x^3}$

Step 1: $(5x^3z^4y^{-2})^{-2}=5^{-2}x^{-6}z^{-8}y^4=\frac{y^4}{25x^6z^8}$.
Step 2: $(3xy^4z^{-3})^3=27x^3y^{12}z^{-9}=\frac{27x^3y^{12}}{z^9}$.
Step 3: Multiply: $\frac{27x^3y^{16}}{25x^6z^{17}}=\frac{27y^{16}}{25z^{17}x^3}$.

19 **Answer: A** $\frac{a\sqrt{b}+b\sqrt{a}}{ab}$

Step 1: For $a,b>0$: $\sqrt{a^2b}=a\sqrt{b}$ and $\sqrt{ab^2}=b\sqrt{a}$.
Step 2: So the expression is $\frac{a\sqrt{b}+b\sqrt{a}}{ab}$.

20 **Answer: C** 3.6

Step 1: Small cylinder: $r=3$, $h=10$: $V=\pi(9)(10)=90\pi$.
Step 2: Large cylinder: $r=4$, $h=20$: $V=\pi(16)(20)=320\pi$.
Step 3: $\frac{320\pi}{90\pi}\approx 3.56\approx 3.6$.

21 **Answer: C** 20%

Step 1: Radii: 0.5, 1, 1.5. Area inside the large circle but outside the small ones: $\pi(1.5)^2-\pi(0.5)^2-\pi(1)^2=\pi(2.25-0.25-1)=\pi$.
Step 2: Square area $=4^2=16$.
Step 3: $P=\frac{\pi}{16}\approx 0.196\approx 20\%$.

22 **Answer: A** 1 ft 1.5 in

Step 1: Convert to inches: hoop 13 ft $=156$ in, its shadow 3 ft 4 in $=40$ in, boy 4 ft 5 in $=53$ in.
Step 2: Same sun angle, so heights and shadows are proportional: $\frac{156}{40}=\frac{53}{s}$.
Step 3: $s=\frac{53\times 40}{156}\approx 13.6$ in ≈ 1 ft 1.5 in.

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23 **Answer: B** 11

Step 1: Adding 5 to each number adds 5 to the mean.
Step 2: $6 + 5 = 11$.

24 **Answer: B** Q_2, Q_4

Step 1: y-intercept 0: the line passes through the origin.
Step 2: Negative slope: going right it goes down, so it lies in quadrant II (left, up) and quadrant IV (right, down).

25 **Answer: A** $(3, 5), (4, 2), (1, 6)$

Step 1: Reflecting across the y-axis changes the sign of x: $(x, y) \to (-x, y)$.
Step 2: $(-3, 5) \to (3, 5)$, $(-4, 2) \to (4, 2)$, $(-1, 6) \to (1, 6)$.

26 **Answer: A** Ice cream

Step 1: Charlie chose cake and Marco had the same, so Marco: cake. Rob: brownies.
Step 2: No one else had what Rob or Lauren had, so Lauren's dessert is unique: not cake (Marco, Charlie), not brownies (Rob), and not cookies (given).
Step 3: Lauren ate ice cream.

27 **Answer: B** x

Step 1: Factor: $x^2 - 100 = (x - 10)(x + 10)$, $x^2 + 4x - 60 = (x + 10)(x - 6)$, $x^2 - 6x = x(x - 6)$.
Step 2: $\frac{(x-10)(x+10)}{(x+10)(x-6)} \cdot \frac{x(x-6)}{x-10}$.
Step 3: Everything cancels except x.

28 **Answer: C** Mean = Median

Step 1: Sum: $75 + 70 + 80 + 70 + 80 + 90 + 95 + 70 + 70 + 75 + 80 + 80 + 90 + 95 = 1120$; mean $= \frac{1120}{14} = 80$.
Step 2: Ordered: $70, 70, 70, 70, 75, 75, \mathbf{80}, \mathbf{80}, 80, 80, 90, 90, 95, 95$; median $= \frac{80+80}{2} = 80$.
Step 3: Mean = Median.

29 **Answer: A** $(1, 5), (5, 4), (12, 6), (14, 6)$

Step 1: Write each fact as (day, hours worked).
Step 2: Day 1: 5 h; day 5: 4 h; days 12 and 14: 6 h.
Step 3: $(1, 5), (5, 4), (12, 6), (14, 6)$.

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30 **Answer: C** 3072

Step 1: Doubling each year: $N = 6 \cdot 2^t$.
Step 2: After 9 years: $6 \cdot 2^9 = 6 \cdot 512 = 3072$.

31 **Answer: D** 5

Step 1: Radii are $5k$, $8k$, $9k$; the middle one is 10: $8k = 10 \Rightarrow k = 1.25$.
Step 2: Smallest radius $= 6.25$, largest $= 11.25$.
Step 3: Distance between the circles $= 11.25 - 6.25 = 5$ cm.

32 **Answer: D** 108

Step 1: Along the 60 cm side: $\frac{60}{5} = 12$ squares; along the 45 cm side: $\frac{45}{5} = 9$.
Step 2: Total: $12 \times 9 = 108$ squares.

33 **Answer: D** 16 to 1

Step 1: Side lengths are in ratio $4 : 2 : 1$ (largest : middle : smallest).
Step 2: Areas go with the square of the side: $4^2 : 1^2 = 16 : 1$.

34 **Answer: C** $ab < 0$ and $a + b > 0$

Step 1: Expand: $(x + a)(x + b) = x^2 + (a + b)x + ab$.
Step 2: Compare: $a + b = m > 0$ and $ab = n < 0$.
Step 3: So $ab < 0$ and $a + b > 0$.

35 **Answer: D** I, II, and III

Step 1: Two squares aligned the same way can overlap in a smaller square or a non-square rectangle.
Step 2: If one square is rotated $45°$ and its corner covers the other's corner, the overlap can be a triangle.
Step 3: So I, II and III are all possible.

36 **Answer: C** It is not a square

Step 1: "If a block is square, then it is not blue" has the contrapositive "if a block is blue, then it is not square".
Step 2: So a blue block must not be a square. (Being blue doesn't force it to be triangular, circular, or non-plastic.)

37 **Answer: A** $m\angle BAD = 105°$ and $m\angle BCD = 75°$

Step 1: $\overline{AC}$ is a diameter, so $\angle ABC = \angle ADC = 90°$.
Step 2: In $\triangle ABC$: $\angle BAC = 90° - 40° = 50°$, so $\angle BAD = 50° + 55° = 105°$.
Step 3: Opposite angles of a cyclic quadrilateral are supplementary: $\angle BCD = 180° - 105° = 75°$.

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38 **Answer: B** -4

Step 1: $P = (x, 3)$. Distance to $(8, 8)$: $(x-8)^2 + (3-8)^2 = 13^2$.
Step 2: $(x-8)^2 = 169 - 25 = 144 \Rightarrow x - 8 = \pm 12$.
Step 3: $x = 20$ or $x = -4$; the one less than 8 is -4.

39 **Answer: C** $\frac{5}{3}$

Step 1: Solve for y: $-3y = -5x + 9 \Rightarrow y = \frac{5}{3}x - 3$.
Step 2: The slope is $\frac{5}{3}$.

40 **Answer: A** 7

Step 1: $5(P + Q) = 30 \Rightarrow P + Q = 6 \Rightarrow P = 6 - Q$.
Step 2: $P > 0$ means $6 - Q > 0$, i.e. $Q < 6$.
Step 3: Q cannot be 7.

41 **Answer: D** Symmetric Property of Equality

Step 1: Switching the two sides of an equation ($a = b \Rightarrow b = a$) is the Symmetric Property of Equality.

42 **Answer: C** The conditional statement and its contrapositive

Step 1: A conditional and its contrapositive always have the same truth value (as do the converse and the inverse).
Step 2: So the conditional statement and its contrapositive are logically equivalent.

43 **Answer: D** Graph D

Step 1: A function gives each input x exactly one output y — the vertical line test.
Step 2: A: vertical lines fail. B: a circle fails. C: the sideways V fails (two y-values for one x).
Step 3: D passes: every vertical line meets it once.

44 **Answer: A** AA

Step 1: $\overline{AB} \parallel \overline{CD}$ gives congruent alternate interior angles: $\angle A \cong \angle D$ and $\angle B \cong \angle C$.
Step 2: The vertical angles where the segments cross are also equal.
Step 3: Two pairs of congruent angles prove similarity: AA.

45 **Answer: A** If Chris doesn't get cavities, he uses toothpaste X.

Step 1: Original: "If p (Chris uses toothpaste X), then q (he doesn't get cavities)."
Step 2: Converse: swap them — "If q, then p".
Step 3: "If Chris doesn't get cavities, he uses toothpaste X."

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46 **Answer: B** $4\sqrt{5}$

Step 1: The radius is $4 + 2 = 6$; the chord is 4 units from the center.
Step 2: A perpendicular from the center bisects the chord: half-chord $= \sqrt{6^2 - 4^2} = \sqrt{20} = 2\sqrt{5}$.
Step 3: $GB = 2 \times 2\sqrt{5} = 4\sqrt{5}$.

47 **Answer: D** Graph D

Step 1: Boundary line $6A + 3S = 1500$ has a negative slope and is solid (because of $\geq$).
Step 2: Test the origin: $0 \geq 1500$ is false, so shade the side away from the origin (above the line).
Step 3: Numbers of people can't be negative, so the region stays in the first quadrant: graph D.

48 **Answer: A** Graph A

Step 1: $c < 0$: the parabola crosses the y-axis below the origin.
Step 2: The vertex is at $x = -\frac{b}{2}$; with $b > 0$ this is negative, so the vertex is left of the y-axis. The x^2 coefficient is 1, so it opens up.
Step 3: Graph A matches all three features.

49 **Answer: C** Infinitely many solutions

Step 1: Rewrite the second equation: $10y = -4x + 14 \Rightarrow 4x + 10y = 14$.
Step 2: Divide by 2: $2x + 5y = 7$ — the same as the first equation.
Step 3: Identical lines: infinitely many solutions.

50 **Answer: C** Graph C

Step 1: The vertex is at $x = -\frac{b}{2a} = -\frac{8}{4} = -2$.
Step 2: $y = 2(4) + 8(-2) + 7 = 8 - 16 + 7 = -1$, so the vertex is $(-2, -1)$; $a = 2 > 0$, so it opens up.
Step 3: Graph C has its vertex at $(-2, -1)$.

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